

# Combinations — Solutions

## Warm-up

Question 1:

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Question 2:

$${}^{10}C_3 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$$

Question 3:

$${}^9C_2 = \frac{9 \times 8}{2 \times 1} = 36$$

Question 4:

$$\text{Ways} = {}^8P_2 = 8 \times 7 = 56$$

Question 5:

$$\frac{n!}{(n-1)!} = \frac{n \times (n-1)!}{(n-1)!} = n$$

Question 6:

$$\text{Ways} = 4 \times 6 = 24$$

Question 7:

$${}^9C_3 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

Question 8:

$${}^{12}C_5 = \frac{12 \times 11 \times 10 \times 9 \times 8}{5 \times 4 \times 3 \times 2 \times 1} = 792$$

Question 9:

$$\text{Ways} = {}^9C_3 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

Question 10:

$$\frac{(n+1)!}{n!} = \frac{(n+1) \times n!}{n!} = n+1$$

Question 11:

$${}^8C_3 = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

Question 12:

$${}^9C_3 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

## Skill-building

Question 1:

$$\text{Ways} = {}^5C_2 \times {}^7C_2 = 10 \times 21 = 210$$

Question 2:

$$\begin{aligned}\frac{(n+1)!}{n!} &= 8 \\ n+1 &= 8 \\ n &= 7\end{aligned}$$

Question 3:

$$\text{Ways} = (6-1)! = 5! = 120$$

Question 4:

$$\text{Case 1} = {}^6C_4 \times {}^4C_1 = 15 \times 4 = 60$$

$$\text{Case 2} = {}^6C_5 \times {}^4C_0 = 6 \times 1 = 6$$

$$\text{Total} = 60 + 6 = 66$$

Question 5:

$$\text{Probability} = \frac{{}^3C_2}{{}^{10}C_2} = \frac{3}{45} = \frac{1}{15}$$

Question 6:

$${}^5P_3 = 5 \times 4 \times 3 = 60$$

Question 7:

$$\text{Ways} = {}^9C_3 = 84$$

Question 8:

$$\text{(a)} = {}^{10}C_5 = 252$$

$$\text{(b)} = {}^6C_3 \times {}^4C_2 = 20 \times 6 = 120$$

$$\text{(c)} = {}^4C_4 \times {}^6C_1 = 1 \times 6 = 6$$

Question 9:

$$\text{(a)} = {}^9C_3 = 84$$

$$\text{(b)} = {}^{10}C_4 - {}^8C_2 = 210 - 28 = 182$$

Question 10:

$$\text{Total} = {}^{10}C_6 \times {}^6C_1 = 210 \times 6 = 1260$$

Question 11:

$$\text{Total} = {}^7C_3 \times 3! = 35 \times 6 = 210$$

Question 12:

$${}^7C_3 = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$$

## Easier Exam Questions

Question 1:

$$\begin{aligned} \text{(i)} &= {}^9C_4 = 126 \\ \text{(ii)} &= 126 - ({}^6C_2 \times {}^3C_2) = 126 - 45 = 81 \end{aligned}$$

Question 2:

$$\begin{aligned} \text{i)} &= {}^{15}C_{10} = 3003 \\ \text{ii)} &= {}^7C_6 \times {}^8C_4 = 7 \times 70 = 490 \\ \text{iii)} &= \frac{490 + ({}^7C_7 \times {}^8C_3)}{3003} = \frac{546}{3003} = \frac{2}{11} \end{aligned}$$

Question 3:

$$\text{Ways} = 2^7 - {}^7C_0 - {}^7C_1 = 128 - 1 - 7 = 120$$

Question 4:

$$\text{Total} = {}^{17}C_8 \times {}^5C_1$$

Question 5:

$$\begin{aligned} \text{(i)} &= {}^{14}C_5 = 2002 \\ \text{(ii)} &= \frac{{}^9C_3 \times {}^5C_2}{2002} = \frac{840}{2002} = \frac{60}{143} \\ \text{(iii)} &= \frac{4! \times 2!}{5!} = \frac{48}{120} = \frac{2}{5} \end{aligned}$$

Question 6:

$$\text{Total} = {}^{16}C_7 \times 6! = \frac{16!}{7!9!} \times 6! = \frac{16!}{7 \times 9!}$$

Question 7:

$$\begin{aligned} \text{(i)} &= {}^3C_2 \times {}^4C_2 = 18 \\ \text{(ii)} &= {}^2C_1 \times {}^3C_1 = 6 \end{aligned}$$

Question 8:

$$\begin{aligned} \text{(i)} &= {}^{12}C_7 = 792 \\ \text{(ii)} &= \frac{{}^{12}C_4 \times {}^8C_4 \times {}^4C_4}{3!} = 5775 \\ \text{(iii)} &= 12 \times 11 = 132 \end{aligned}$$

Question 9:

$$\begin{aligned} \text{(i)} &= {}^{14}C_5 = 2002 \\ \text{(ii)} &= (84 \times 10) + (126 \times 5) + (126 \times 1) = 1596 \\ \text{(iii)} &= 2002 - {}^{12}C_3 = 2002 - 220 = 1782 \end{aligned}$$

Question 10:

$$\text{Total} = {}^{12}C_8 \times {}^8P_5 = \frac{12!}{8!4!} \times \frac{8!}{3!} = \frac{12!}{4!3!}$$

Question 11:

$$(b) = {}^{10}C_3 + ({}^{10}C_1 \times {}^{10}C_2) = 120 + 450 = 570$$

$$(c) = 16(n-1)! = 5n! + (n+1)!$$

$$16 = 5n + n(n+1)$$

$$n^2 + 6n - 16 = 0 \implies n = 2$$

## Harder Exam Questions

Question 1:

$$\text{i)} = {}^5C_3 \times {}^6C_1 = 60$$

$$\text{ii)} = \frac{{}^4C_2 \times {}^5C_0}{60} = \frac{6}{60} = \frac{1}{10}$$

Question 2:

$$\text{Ways} = {}^8C_6 - {}^6C_4 = 28 - 15 = 13$$

Question 3:

$$\text{(i)} = 23 \times 22 = 506$$

$$\text{(ii)} \alpha = {}^{12}C_2 \times {}^{11}C_2 = 3630$$

$$\text{(ii)} \beta = \frac{{}^{11}C_1 \times {}^{10}C_1}{3630} = \frac{110}{3630} = \frac{1}{33}$$

Question 4:

$$\text{(i)} = ({}^6C_3 \times {}^7C_2) + ({}^6C_4 \times {}^7C_1) + ({}^6C_5 \times {}^7C_0) = 531$$

$$\text{(ii)} = \frac{({}^5C_3 \times {}^6C_1) + ({}^5C_2 \times {}^6C_2)}{420} = \frac{210}{420} = \frac{1}{2}$$

Question 5:

$$\text{(i)} = {}^{12}C_7 = 792$$

$$\text{(ii)} = \frac{{}^7C_4 \times {}^5C_3}{792} = \frac{350}{792} = \frac{175}{396}$$

Question 6:

$$\text{i.} = {}^8C_3 = 56$$

$$\text{ii.} = \frac{{}^4C_3}{56} = \frac{4}{56} = \frac{1}{14}$$

Question 7:

$$\text{(i)} = 531$$

$$\text{(ii)} = 1/2$$

**Question 8:**

$$(a) = 15 \times 14 = 210$$

$$(b) = {}^{15}C_4 = 1365$$

$$(c) = {}^8C_2 \times {}^7C_2 = 588$$

**Question 9:**

$$\text{Sum} = 60(1000 + 100 + 10 + 1) = 66660$$

**Question 10:**

$$n^2 + 6n - 16 = 0 \implies n = 2$$

**Question 11:**

$$c) (i) = \frac{20!}{12!6!2!}$$

$$(ii) = {}^{10}C_2 \times \frac{18!}{12!6!} = 835380$$

$$d) = \frac{30 + 15 + 1}{56} = \frac{46}{56} = \frac{23}{28}$$

**Question 12:**

$$(a) = \frac{-n^2 - n + 1}{(n + 1)!}$$

$$(c) i. = {}^{17}C_5 = 6188$$

$$(ii) = \frac{{}^9C_5}{{}^{17}C_5} = \frac{63}{3094}$$

$$(e) = {}^{10}C_5 - {}^8C_3 = 196$$

**Question 13:**

$$(i) = {}^nC_{r-1}$$

$$(ii) = {}^{101}C_3 - {}^3C_3 = 166649$$

**Question 14:**

$$(i) = {}^6C_4 \times {}^5C_3 = 150$$

$$(ii) = 2! \times 4! \times 3! = 288$$

**Question 15:** (i) There are 6 dots and any 3 points can be chosen to form a triangle. The order of the 3 points doesn't matter as the same 3 points form the same triangle. Therefore there are  ${}^6C_3$  distinct triangles.

(ii) There are a couple of ways this can be done.

One way is to take all groups of 3 points ( ${}^{10}C_3$ ) and subtract all groups of 3 on each row ( ${}^5C_3 \times 2$  for 2 rows), which means there are  ${}^{10}C_3 - 2 \times {}^5C_3 = 100$  distinct triangles.

Another way is to select 2 dots from one row and one dot from the other. Lets say we pick 2 dots from the top row and 1 dot from the bottom. That results in  ${}^5C_2 \times {}^5C_1 = 50$  distinct triangles. We will get another 50 triangles by selecting 2 dots on the bottom row and 1 dot from the top row (same logic as before but upside-down), resulting in 100 distinct triangles again.

**Question 16:** The group can have one person, 2 people, 3 people and so on, up to 22 people (there can't be all 23 since that leaves the second group empty)

There are  $\binom{23}{1}$  ways to form a group of 1 person.

There are  $\binom{23}{2}$  ways to form a group of 2 people.

There are  $\binom{23}{3}$  ways to form a group of 3 people. (Hopefully you get the idea)

so the total number of ways to form a group of 1 person or 2 people or 3 people up until 22 people are

$$\binom{23}{1} + \binom{23}{2} + \binom{23}{3} + \dots + \binom{23}{22}$$

Consider the given statement in the question:  $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$   
Letting  $n = 23$ , we get the closest match to our desired sum:

$$\binom{23}{0} + \binom{23}{1} + \binom{23}{2} + \dots + \binom{23}{23} = 2^{23}$$

But we just need to remove  $\binom{23}{0}$  and  $\binom{23}{23}$ . Each of them are equal to 1, so our final number of ways simplifies to:

$$\text{Ways} = 2^{23} - 2$$

**Question 17:**

(i) Choosing 3 points out of  $n$ :  ${}^nC_3$

(ii) There are  $2n - 1$  points (being careful not to double count point X). We take all groups of 3 points ( ${}^{2n-1}C_3$ ), and remove the number of ways to choose 3 points on each line ( ${}^nC_3$  for each line so a total of  $2 \times {}^nC_3$  for both), so the total number of triangles are  ${}^{2n-1}C_3 - 2 \times {}^nC_3$

(iii) ( $\alpha$ ) We have to choose 1 point out of  $n - 1$  remaining points on  $l_1$  and 1 point out of  $n - 1$  remaining points on  $l_1$ . This simplifies to  $(n - 1)^2$

(iii) ( $\beta$ ) We have to select 2 points on line  $l_1$  and one point on line  $l_2$ . On  $l_1$  there are  $n$  points and we are choosing 2 of them, so that is  ${}^nC_2$ . On  $l_2$  there are  $n - 1$  remaining points (can't select point X since that is not a triangle, and also already selected in  $l_1$ ), and we are choosing 1, so that is  ${}^{n-1}C_1$ .

Altogether that is  ${}^nC_2 \times {}^{n-1}C_1$  triangles that have 2 vertices on  $l_1$ . There is also the same number of triangles that have 2 vertices on  $l_2$  (we will need this for part (iv))

(iv) The difficulty in this question lies in correlating the previous results.  
We are required to show that:

$${}^{2n-1}C_3 = 2 \times {}^nC_3 + (n - 1)^3$$

Notice 2 of the terms in the above equation match the solution in (ii), namely  ${}^{2n-1}C_3$  and  $2 \times {}^nC_3$ . Since (ii) asked for the total number of possible triangles, one could correlate parts (ii) and (iii) (with a lot of imagination) with the statement below:

"The total number of triangles either have 2 vertices on  $l_1$  or 2 vertices on  $l_2$ , but since X is on both lines, we double counted all triangles that have X as a vertex"

As an equation, substituting the results in (ii) and (iii), we get:

$${}^{2n-1}C_3 - 2 \times {}^nC_3 = 2 \times {}^nC_2 \times {}^{n-1}C_1 - (n-1)^2$$

All we need to do now is show that the right side of this is equal to  $(n-1)^3$ .

$$\begin{aligned} {}^nC_2 &= \frac{n!}{2!(n-2)!} \\ &= \frac{(n)(n-1)}{2} \\ \text{our other term } {}^{n-1}C_1 &= \frac{(n-1)!}{1!(n-1-1)!} \\ &= \frac{(n-1)!}{(n-2)!} \\ &= n-1 \end{aligned}$$

Putting all these terms together:

$$\begin{aligned} {}^{2n-1}C_3 - 2 \times {}^nC_3 &= 2 \times {}^nC_2 \times {}^{n-1}C_1 - (n-1)^2 \\ &= 2 \times \frac{(n)(n-1)}{2} \times (n-1) - (n-1)^2 \\ &= n(n-1)^2 - (n-1)^2 \\ &= (n-1)^2(n-1) \\ &= (n-1)^3 \\ \therefore {}^{2n-1}C_3 &= 2 \times {}^nC_3 + (n-1)^3 \end{aligned}$$